

SHUBHAM KULKARNI

AI Engineer | Computer Vision · Edge AI · Generative AI & RAG · MLOps

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PROFESSIONAL SUMMARY

AI Engineer who designs and ships production computer vision and generative AI systems for national highway infrastructure. Core engineer on **VIDES**, an edge traffic-enforcement network on NVIDIA Jetson (14 detection types, 7 models per node, 50,000+ events/day at 95% accuracy, sub-40 ms frame processing), and **TMCS**, an NHAI ATMS-aligned incident-detection platform with court-grade evidence. Owns systems end to end — TensorRT model optimization, multi-object tracking, real-time streaming, data pipelines, access control and CI/CD — and builds RAG, hybrid-search and multi-agent platforms. National Champion (AIR 1), Smart India Hackathon 2023.

CORE COMPETENCIES

Computer Vision & Deep Learning: Object Detection (YOLOv5/v8, RT-DETR), Multi-Object Tracking (ByteTrack, Kalman Filter), Re-Identification (OSNet-AIN, ResNet), ANPR & OCR (PaddleOCR PP-OCRv4), OpenCV, PyTorch, TensorFlow, Hugging Face, NLP

Edge AI & Inference Optimization: NVIDIA Jetson (Orin NX/Nano, Xavier), TensorRT (FP16/INT8), CUDA, ONNX Runtime, GStreamer (nv4l2decoder), Hardware-Accelerated Video Decoding, Model Quantization, Raspberry Pi, Rockchip

Generative AI & Agents: Retrieval-Augmented Generation (RAG), Hybrid Search (FAISS + BM25 + RRF), LangChain, Multi-Agent Orchestration, Google ADK, Model Context Protocol (MCP), Prompt Engineering, Knowledge Graphs, Vector Databases (FAISS, ChromaDB), Gemini, Mistral, Ollama

Backend, Streaming & Systems: Python, FastAPI, Flask, Django/DRF, AsyncIO, asyncpg, REST APIs, Microservices, Event-Driven Architecture, Server-Sent Events, WebSockets (Socket.IO), RTSP, ONVIF, MediaMTX, Multithreading & Concurrency

MLOps, Cloud & Data: Docker, Kubernetes, CI/CD, Git, Linux, systemd, AWS, Microsoft Azure, GCP (Vertex AI), PostgreSQL, Redis, MongoDB, MySQL, RBAC, Automated Testing

PROFESSIONAL EXPERIENCE

Software Engineer — AI/ML | Arya Omnitalk Wireless Solutions Pvt. Ltd. (Arvind Group), Pune **Aug 2025 – Present**

VIDES — Edge Traffic Enforcement & ATCC Network | NVIDIA Jetson Orin, TensorRT, GStreamer, YOLOv8, RT-DETR, PaddleOCR

- **Architected the multi-model edge pipeline:** ran 7 models (6 TensorRT FP16 engines + PP-OCRv4) concurrently on one Jetson to deliver 14 detection types — ANPR, helmet, triple riding, seatbelt, speed, wrong-way, stopped vehicle, accident, pedestrian, animal and pothole — at 50,000+ events/day and 95% accuracy.
- **Removed the CPU bottleneck:** built GPU-direct ingestion with GStreamer and nv4l2decoder plus a tee-based H.264 passthrough (no re-encode on the encoder-less Orin Nano), sustaining sub-40 ms frame processing across 6 concurrent 1080p RTSP streams.
- **Designed a fan-out/fan-in decision layer:** parallel ANPR, helmet and seatbelt worker pools on bounded queues, fused per vehicle by a detection finalizer with a 6 s timeout, so no single slow model can stall a record.
- **Automated highway toll auditing:** built the ANPR path — YOLO plate localization, CLAHE preprocessing, Laplacian blur penalties and 2-read OCR consensus — logging every vehicle passage with evidence and eliminating manual reporting.
- **Delivered sub-150 ms operator alerting:** decoupled inference from the UI with an async DB writer and an SSE → Socket.IO relay (exponential-backoff reconnect, rolling per-camera cache), with human-in-the-loop verification of every violation.

TMCS — Highway Incident Detection (NHAI ATMS 2023) | YOLOv8, RT-DETR, ByteTrack, FastAPI, PostgreSQL, ONVIF

- **Built the detection and rule engine:** dual-model detection (YOLOv8 + RT-DETR) with ByteTrack and a dt-aware Kalman filter across 7 incident rules and 11 object classes; 1.5 ms p50 tracking-and-rules latency (10.3 ms at 80 objects) against a p95 ≤ 45 ms inference target.
- **Cut false alarms at the source:** rider-aware occupant tagging reduced false pedestrian alarms by 95%+; a night-mode glare guard (EMA luma + HSV filtering) and PTZ-preset-aware analytics prevent false events at night and during camera movement.
- **Hardened the device and data layers:** ONVIF circuit breaker rejects dead cameras in 0.1 ms (vs 10–30 s timeouts); preset sync cut from 2,500 ms to under 10 ms (−99.6%); TTL caching cut analytics SQL load by 90%+; deterministic shutdown in under 0.5 s.
- **Made evidence court-ready:** write-once evidence with SHA-256 integrity and full model/rule/config version traceability; RBAC with 36 permissions across 5 role tiers; 298+ automated unit, integration and fault-injection tests.

Vehicle Re-Identification & Forensic Matching | *PyTorch, YOLOv5, OSNet-AIN, FAISS IVF, FastAPI, Docker*

- **Owned the forensic search service:** designed a stateless, database-free re-identification microservice using 512-dimensional OSNet-AIN embeddings and FAISS IVF indexing — sub-100 ms search across 150,000+ daily forensic queries, 30× faster retrieval.
- **Ran production operations:** CI/CD, heartbeat monitoring and automated recovery kept edge nodes at 99.9% uptime; worked across hardware, backend and operations teams from pilot to production.

Earlier Experience

- **AI Intern, IBM India (Jun – Aug 2023):** built transformer-based NLP text classifiers reaching 90% accuracy on production datasets.
- **AI Intern, MathWorks (May – Sep 2023):** developed deep learning and image-processing solutions in MATLAB for computer vision use cases.
- **Data Analytics Trainee, AICTE · VOIS for Tech · Vodafone Idea Foundation (Oct – Dec 2024):** applied LLMs to agriculture and science analytics pipelines.

SELECTED PROJECTS

CivicMind — Smart City Decision Intelligence | *Gemini 2.5, Vertex AI, FastAPI, Next.js 14, Docker* **2026**

github.com/kulkarnishub377/CivicMind

- Multi-agent orchestration platform (4 coordinated agents) with a digital-twin simulator and an Explainable AI layer for transparent, risk-based policy recommendations.

OmniSight — Multi-Modal Incident Intelligence | *FastAPI, Agentic AI, RAG, Vector DB, Docker* **2026**

github.com/kulkarnishub377/OmniSight

- Agentic workflow engine that correlates high-velocity IoT, camera and audio signals in real time to validate anomalies and generate safety guardrails.

DocuAI Studio v3.1 — Document AI + RAG Platform | *FastAPI, LangChain, FAISS, Mistral, Ollama* **Mar 2026 – Present**

github.com/kulkarnishub377/Document-AI---RAG-Pipeline

- Locally hosted RAG platform (40+ API endpoints) for PDFs, images and web URLs with sub-2 s answers and zero external data exposure.
- Hybrid retrieval fusing FAISS dense search with BM25 via Reciprocal Rank Fusion to reduce hallucinations; knowledge-graph entity extraction and complexity-based routing between local (Mistral 7B) and cloud LLMs.

JalTantra — Smart Irrigation & AgTech Platform | *Python, IoT, Computer Vision, ML, GenAI* **Dec 2024 – May 2025**

shubhamkulkarni.me/JalTantra

- IoT irrigation with predictive ML and CV crop-disease detection, improving water efficiency by 30%; multilingual (Hindi, Marathi, English) advisory chatbot. University Rank 2, SPPU Startup Olympiad 2025.

Alumni Management Portal — Officially Adopted by the College | *Django DRF, PostgreSQL, WebSockets* **2024 – 2025**

github.com/kulkarnishub377/Alumni_Management_portal

- Role-based platform (19 data models, 50+ REST APIs, 4 dashboards); cut query latency from 3,200 ms to 12 ms with composite indexing and resolved JWT-refresh race conditions with a custom WebSocket subscriber queue.

EDUCATION

B.E., Electronics & Telecommunication — First Class | Savitribai Phule Pune University (SPPU) **Dec 2021 – Mar 2025**

Dr. Vithalrao Vikhe Patil College of Engineering, Ahilyanagar

ACHIEVEMENTS & CERTIFICATIONS

- **AIR 1 — Smart India Hackathon 2023 (National Champion):** team lead; built an AI e-waste classification and routing system, selected best nationally.
- **Runner-Up (Global) — TIAA Global AI Hackathon:** AI investment-intelligence platform with risk-based scoring.
- **University Rank 2 — SPPU Startup Olympiad 2025:** JalTantra precision-agriculture platform.
- **Certifications:** Google Cloud Gen AI Academy APAC 2026 (Hack2skill) · Deploy Multi-Agent Architectures — Google (2026) · Engineer AI Agents with ADK — Google (2026) · Build AI Agents with Enterprise Databases — Google Cloud (2026) · Introduction to Machine Learning Concepts — Microsoft (2026) · Getting Started with AI on Jetson Nano — NVIDIA DLI (2025) · Career Essentials in Generative AI — Microsoft & LinkedIn · Deep Learning & Statistics Onramp — MathWorks